

SALEM 10 W, MO, USA

WMO: 725600

Lat: 37.6344N

Lon: 91.7226W

Elev: 1198

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 23909

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.6	10.2	-4.0	4.6	6.3	0.7	6.0	13.5	14.5	49.1	13.0	45.5	3.1	N/A	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	93.9	73.5	90.6	73.7	88.1	73.3	77.4	87.7	76.3	86.4	75.0	84.8	5.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.5	134.9	83.1	73.2	128.9	82.0	72.0	123.8	80.4	41.7	87.6	40.5	86.2	39.3	85.0	81.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.4	11.9	10.4	DB	-6.3	97.3	9.0	4.3	-12.8	100.4	-18.0	102.9	-23.1	105.2	-29.6	108.3	
			WB	-7.8	79.2	9.4	1.7	-14.6	80.4	-20.1	81.4	-25.4	82.4	-32.2	83.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	31.2	35.1	45.4	55.7	63.6	72.0	75.1	73.7	67.1	55.9	44.3	35.5
	DBStd	17.82	12.04	12.04	11.36	9.20	8.26	5.53	5.43	5.72	7.15	9.49	10.00	10.06
	HDD50	2040	591	438	223	46	7	0	0	0	47	226	462	
	HDD65	4853	1047	836	611	299	128	12	2	5	61	313	623	916
	CDD50	3736	9	22	81	217	427	660	778	736	513	228	54	11
	CDD65	1078	0	0	5	21	84	222	316	275	124	30	1	0
	CDH74	10931	0	3	60	243	825	2201	3273	2768	1271	274	13	0
	CDH80	4168	0	0	7	41	196	780	1443	1205	446	49	1	0
Wind	WSAvg	4.2	4.8	5.2	5.6	6.1	4.7	3.8	2.9	2.8	2.7	3.6	4.3	4.5
Precipitation	PrecAvg	46.2	2.6	2.6	3.9	4.6	5.6	4.1	4.0	3.9	4.4	3.3	4.0	3.0
	PrecMax	60.4	5.7	6.3	10.7	12.5	12.6	8.3	11.1	8.7	12.7	10.6	9.6	9.9
	PrecMin	27.1	0.6	0.4	1.1	0.7	0.9	1.4	1.3	1.7	0.4	0.3	0.7	0.7
	PrecStd	9.4	1.2	1.5	2.0	2.4	2.6	1.9	2.1	1.6	2.9	2.0	2.3	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.7	72.9	80.2	84.7	88.9	94.6	98.6	98.4	91.7	85.3	75.4	66.9
		MCWB	54.5	59.0	61.4	64.0	67.3	73.0	71.7	73.9	72.6	69.1	59.9	57.4
	2%	DB	61.8	66.5	74.7	80.1	84.7	89.2	94.3	94.0	88.5	80.4	70.2	61.9
		MCWB	50.7	54.5	59.2	62.0	68.6	74.3	73.3	74.0	72.6	65.3	56.4	55.3
	5%	DB	55.6	61.5	69.0	76.7	82.0	87.0	90.6	90.4	85.6	77.1	65.7	57.0
		MCWB	47.2	52.5	55.3	60.6	67.8	73.8	74.7	73.2	71.6	63.3	55.7	49.2
	10%	DB	50.3	55.7	64.6	72.7	79.0	84.9	87.8	87.0	81.9	72.5	61.8	52.6
		MCWB	43.7	48.1	53.1	59.1	66.6	72.6	74.2	72.8	69.2	61.1	52.6	46.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	59.2	61.2	64.1	67.9	73.1	78.1	79.0	78.8	75.5	71.2	64.5	60.7
		MCDB	64.4	69.8	76.1	79.7	83.6	87.9	88.7	90.1	86.1	81.3	71.1	65.3
	2%	WB	53.3	58.4	61.2	65.7	71.4	76.5	77.6	77.1	74.1	69.2	60.5	56.5
		MCDB	58.0	65.1	72.0	75.6	80.9	86.6	87.6	87.9	84.7	76.2	66.4	60.6
	5%	WB	48.4	54.2	58.1	63.5	69.9	75.1	76.5	75.5	72.9	66.4	57.0	51.4
		MCDB	53.7	58.6	66.3	72.1	78.7	84.9	86.5	85.5	83.0	72.3	63.5	55.7
	10%	WB	43.7	47.4	55.1	61.4	68.4	73.7	75.2	74.0	71.3	63.8	53.3	47.2
		MCDB	50.2	55.1	61.5	70.4	76.8	82.5	84.9	83.1	80.2	70.9	60.4	50.9
Mean Daily Temperature Range	5% DB	MDBR	21.6	21.5	23.5	23.8	22.2	21.6	22.7	23.1	24.1	24.7	23.3	20.3
		MCDBR	29.2	28.9	29.7	27.8	26.0	24.3	26.1	27.9	26.8	27.2	27.8	27.6
		MCWBR	19.5	20.1	16.2	13.9	12.3	11.8	9.8	10.4	12.4	13.8	17.3	18.2
	5% WB	MCDBR	23.5	25.9	24.6	23.1	21.5	21.3	21.8	22.1	24.0	21.7	22.7	21.7
		MCWBR	18.5	19.9	15.3	13.9	11.5	11.4	10.1	10.1	11.6	12.5	16.7	17.8
Clear-Sky Solar Irradiance	taub	0.299	0.307	0.334	0.382	0.417	0.425	0.454	0.448	0.385	0.343	0.311	0.296	
	taud	2.493	2.464	2.435	2.307	2.262	2.265	2.215	2.226	2.382	2.491	2.529	2.516	
	Ebn at Noon	283	295	295	285	275	272	263	262	274	277	276	276	
	Edh at Noon	25	30	34	40	43	43	45	43	35	29	24	23	
All-Sky Solar Radiation	RadAvg	719	921	1235	1600	1789	2047	1985	1775	1529	1122	780	607	
	RadStd	74	106	110	96	158	150	134	95	98	124	84	70	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (22 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page